

2023



Autodesk Fusion Delivers Generative Design Capabilities Powered by AWS

Learn how Autodesk Fusion software is helping companies achieve breakthroughs in product design and manufacturing on AWS.

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Generates hundreds
of innovative designs quickly

Automates



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Accelerates
design exploration during product dev



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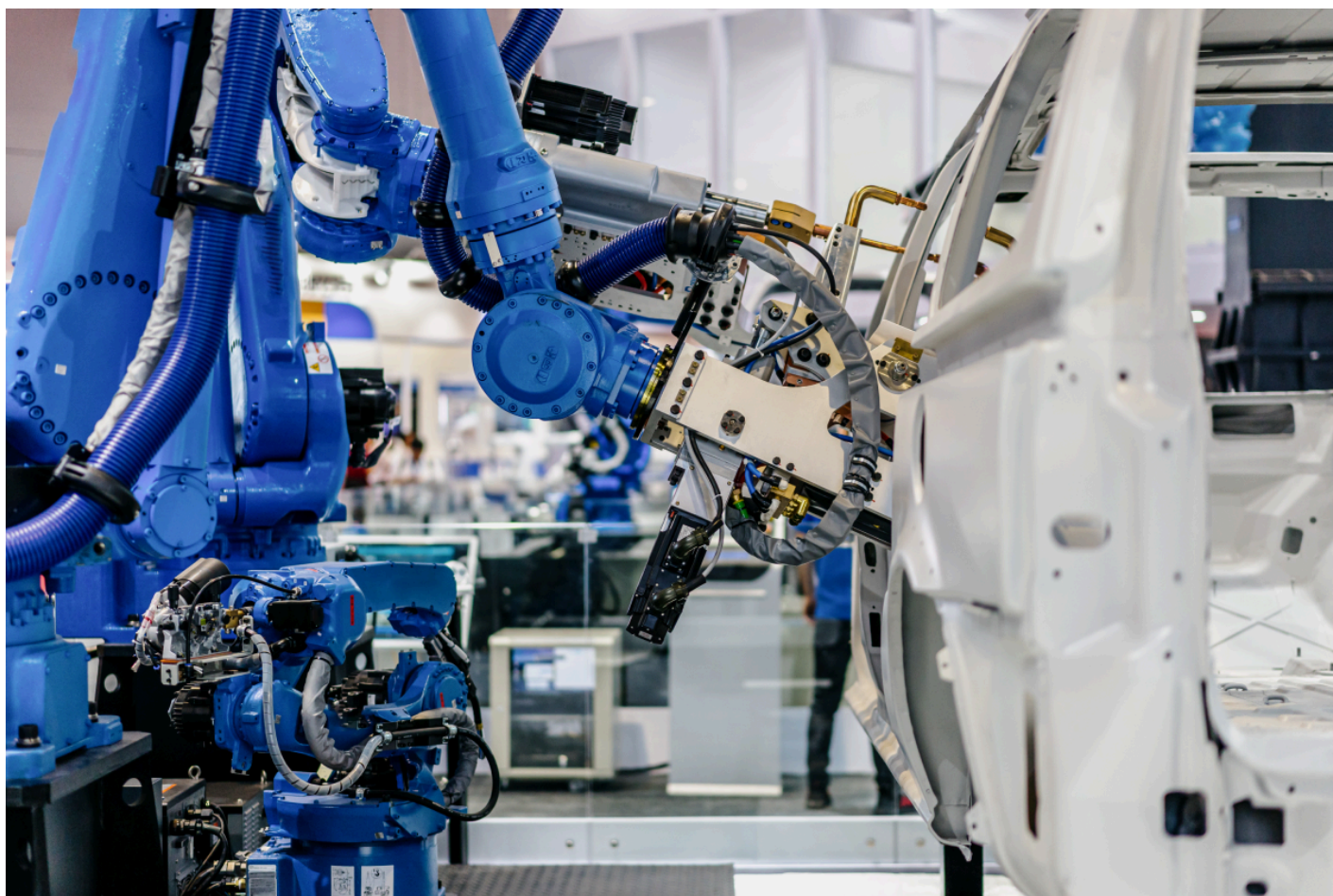


Overview

Product development involves a series of complex decisions and trade-offs, from conceptualization to the final blueprint. [Autodesk](#) Fusion combines design, mechanical engineering, electronics engineering, and manufacturing software in a single, cloud-based offering for a complete product development workflow. It seeks to simplify the product development process by helping designers explore multiple design options with generative design technology.

Generative design requires a robust compute infrastructure capable of handling vast datasets and performing intensive computations in parallel to help its users arrive at the best design in the least amount of time. Turning to Amazon Web Services (AWS), Autodesk tapped into a scalable, high-performance cloud environment to facilitate generative design in Autodesk Fusion software.

The generative design feature processes and analyzes design parameters automatically, offering designers an array of viable solutions based on their specified constraints. With this solution, Autodesk has empowered its customers to achieve entirely new, innovative designs and bring their products to market faster.



Opportunity | Using AWS Services to Bring Generative Design to Life for Autodesk

Founded in 1982, Autodesk software is used by over 100 million people worldwide. Aimed at product design and manufacturing spaces, Autodesk Fusion is an all-in-one tool for product design that empowers teams to bring ideas to realization through advanced data management, 3D modeling, and



electronics design tools. To help its customers work faster and smarter, Autodesk Fusion released its generative design functionality in 2018.

“Generative design was one of our first commercial applications based on our deep research into design automation and optimization,” says Alexander Stern, head of manufacturing platform strategy at Autodesk. “In the past, a lot of the tasks that generative design can handle were done manually and required a lot of time and effort. With this feature, customers can automate workflows and find better outcomes.”

Built from years of intensive exploration, generative design can generate hundreds of different design options based on constraints selected by the user, such as material type, weight, cost, and manufacturing methods. Based on its extensive history working with the AWS team, Autodesk chose AWS once again to serve as the foundation for generative design and to support its go-to-market strategy.

“The goal of generative design is to help accelerate and expand the design exploration phase during product development,” says Mike Smell, senior product manager of Fusion generative design at Autodesk. “The exploration space can become quite large, so a high-performing and scalable compute infrastructure was a critical part of bringing our solution to market. We found the solution on AWS.”

Solution | Helping PIX Moving Quickly Generate Hundreds of Design Options

Generative design is built on top of a common data model foundation that is designed to support parallelized high-speed compute. A cloud-based compute environment carries out these jobs, powered by AWS services such as [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances, which provide secure and resizable compute capacity for virtually any workload. Autodesk uses different Amazon EC2 instances powered by GPUs for design simulation, including [Amazon EC2 G5 Instances](#)—the latest generation of NVIDIA GPU-based instances that can be used for a wide range of graphics-intensive and machine learning use cases—as well as other instance types and CPUs.

To achieve optimal cost savings and performance, Autodesk also adopted a mix of [Amazon EC2 Spot Instances](#), which run fault-tolerant workloads at up to a 90 percent discount for AWS customers, and [Amazon EC2 On-Demand Instances](#), which lets customers pay for compute capacity by the hour or second. To manage the different types of compute that it uses for simulation, Autodesk uses [Amazon Elastic Container Service](#) (Amazon ECS), a fully managed container orchestration service. [AWS Batch](#), a fully managed batch computing service, is used to run simulation jobs on Amazon EC2 instances powered by GPUs.

Users can explore the results produced by generative design in an integrated, web-based dashboard. “Oftentimes, our customers have breakthroughs using this functionality that they otherwise wouldn’t have discovered,” says Stern. “If you apply generative design to different design challenges, there is a lot of potential for optimization.”

PIX Moving, a company that specializes in designing [autonomous mobility devices and vehicles](#), used generative design to create its Robobus. This autonomous driving vehicle is equipped with a customizable interior, which customers can design according to their business needs. PIX Moving primarily markets Robobus to retailers and enterprises with customized needs, with use cases ranging from mobile cafes and yogurt stores to on-demand offices.

Designing Robobus presented unique challenges; PIX Moving needed to find ways to balance structural strength and to optimize the interior of the bus to maximize space. With generative design, they overcame these obstacles and produced Robobus’s interior decoration in just 1–2 months. The product has been deployed with Robobus in Korea, Spain, Japan, India, and the United States.



“Using generative design, we can quickly identify high-performance design options that seamlessly integrate with our manufacturing processes,” says Vicky Zhang, algorithm engineer at PIX Moving. “It has empowered us to push the boundaries of innovation, accelerate product development, and achieve our goals more efficiently on the cloud.”

Outcome | Hosting Generative Design Products in the Cloud to Maximize Efficiency

With the capabilities of generative design, Autodesk empowers its customers to innovate faster. Autodesk’s long-term strategy is to host all of its solutions in the cloud and realize new capabilities, faster speeds, and a lower cost of flexibility for customers.

“For generative design, AWS empowers us to use flexible compute power and explore new technologies,” says Stern. “As we move to the cloud, I expect to see more combinations of AWS services and Autodesk products.”

About Autodesk

Autodesk provides software products and services across several industries to empower innovators to bring their ideas to life. Its Fusion software helps design, engineering, and manufacturing professionals quickly create and iterate on 3D models.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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Amazon EC2 G5 Instances

Amazon EC2 G5 instances are the latest generation of NVIDIA GPU-based instances that can be used for a wide range of graphics-intensive and machine learning use cases.

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AWS Batch

AWS Batch lets developers, scientists, and engineers efficiently run hundreds of thousands of batch and ML computing jobs while optimizing compute resources, so you can focus on analyzing results and solving problems.

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Amazon ECS

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